

Module 08: Final - Debugging Assignment

In this assignment you will demonstrate your knowledge of debugging by fixing the errors you find in the program below. Fix the code, and paste it in this document, along with a list of the problems you fixed.

This example allows the user to create an ArrayList that contains a list of states. There should be only one entry per state in the list. There is another method to verify the uniqueness of the list.

File TestStoreStrings.java

```
import java.util.ArrayList;
import stringsCollections.StoreStrings;

public class TestStoreStrings {
    public static void main(String[] args) {
        ArrayList<> states = new ArrayList<Integer>();
        states.add("Virginia");
        states.add("New York");
        states.add("Michigan");
        states.add("Indiana");
        states.add("Missouri");
        states.add("Indiana");
        states.sort(null);

        for(String state:states) {
            System.out.println(state)
        }

        if (StoreStrings.duplicateString(states)) {
            System.out.println("Error! Contains more than one entry of the same state in the
list. Must have only one entry for the state in the list");
        }
        else {
            System.out.println(StoreStrings.count(states));
        }
    }
}
```

File StoreStrings:

```
package stringCollections;
import java.util.ArrayList;

public class StoreStrings {

    public static int count(ArrayList <Integer> words) {
        int lSize = 1;
        for(String i : words) {
            lSize++;
        }

        return lSize;
    }

    public static boolean duplicateString(ArrayList <String> words) {
        boolean isDuplicateString = False;

        ArrayList<> items = new ArrayList<String>();

        for(String word : words) {
            if (items.contains(word)) {
                isDuplicateString = False;
            }
            else {
                items.add(word);
            }
        }

        return isDuplicateString;
    }
}
```

For student to complete:

CORRECTED CODE:

TestStoreStrings:

```
import java.util.ArrayList;

import stringsCollections.StoreStrings;

public class TestStoreStrings {

    public static void main(String[] args) {

        ArrayList<String> states = new ArrayList<>();

        states.add("Virginia");
        states.add("New York");
        states.add("Michigan");
        states.add("Indiana");
        states.add("Missouri");

        states.sort(null);

        for(String state:states) {

            System.out.println(state);

        }

        if (StoreStrings.duplicateString(states)) {

            System.out.println("Error! Contains more than one entry of the same state in the
list. Must have only one entry for the state in the list");

        } else {

            System.out.println(StoreStrings.count(states));

        }

    }

}
```

```
}
```

StoreStrings:

```
package stringsCollections;

import java.util.ArrayList;

public class StoreStrings {

    public static int count(ArrayList <String> words)
    {
        int lSize = 0;
        for(String i : words)
        {
            lSize++;
        }

        return lSize;
    }

    public static boolean duplicateString(ArrayList <String> words) {
        boolean isDuplicateString = false;

        ArrayList<String> items = new ArrayList<>();

        for(String word : words)
        {
            if (items.contains(word)) {
```

```

        isDuplicateString = true;

    }else {

        items.add(word);

    }

}

return isDuplicateString;

}

}

```

LIST OF PROBLEMS FIXED:

TestStoreStrings:

- Changed ArrayList<> states = new ArrayList<Integer>(); to ArrayList<String> states = new ArrayList<>();
- Added semicolon on line 9
- Removed line 13
- Added semicolon on line 18 (now 17)

StoreStrings:

- Changed package stringCollections; to package stringsCollections;
- Changed public static int count(ArrayList <Integer> words) { to public static int count(ArrayList <String> words) {
- Changed int lSize = 1; to int lSize = 0;
- Changed isDuplicateString = False; to isDuplicateString = false;
- Changed ArrayList<> items = new ArrayList<String>(); to ArrayList<String> items = new ArrayList<>();
- Changed isDuplicateString = True; to isDuplicateString = true;